

Reading Eight Erdős Problems Together

A working record: proofs, eliminations, rediscoveries and open questions

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ABSTRACT

This record supports the note [1]. It proves the two lemmas about base 2 that the note uses without proof, lists the negative results of the seven companion notes with their locations and status, gives in full the misreadings of one computation, the fixed-depth theorem and an exact late rejection, records the ideas that were eliminated together with the statement that eliminated each, records what turned out to be known already, and ends with the questions we think are worth someone's time. It is a lookup document. Each fact has one home here.

1 What the note relies on

In the note	Here
Theorem 1.1 (choices against contraction)	Proved in the note. Attribution in Section 6.
Theorem 1.2(b), base 2	Quoted from [23, Section 7]. Removed intervals: Lemma 2.1.
Theorem 1.3 (chains)	Proved in the note. A wrong citation caught on the way: Section 5, item 4.
Theorem 1.4 (fixed-depth rational count)	Proved in the note. Its limits and an exact late rejection: Section 4.
Limits on methods	Section 3, with status and sources.

Every proof here is an ordinary proof. Nothing is checked in Lean. The computations are exact rational arithmetic with saved outputs [2].

2 Two lemmas at base two

Notation is that of the note at $t = 2$: $w_n = (2^n - 1)^{-1}$, $R_N = \sum_{n>N} w_n$, $g_n = w_n - R_n > 0$, $X_F = \sum_{n \in F} w_n$ for finite F , and $\mathcal{A}$ is the set of all subsums.

* *Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Lemma 2.1 (the removed intervals). $[0, E] \setminus \mathcal{A}$ is the disjoint union, over finite nonempty F , of the open intervals $(X_F - g_{\max F}, X_F)$. Their total length is $\sum_n 2^{n-1} g_n = E - 1$, and $g_n = \sum_{j \geq 2} \frac{2^j - 2}{2^j - 1} 2^{-jn} = \frac{2}{3} 4^{-n} + \frac{6}{7} 8^{-n} + \dots$.

Proof. Fix the digits $\varepsilon_1, \dots, \varepsilon_{n-1}$ and let $s = \sum_{i < n} \varepsilon_i w_i$. The values with these digits lie in $[s, s + R_{n-1}]$, and they split into $[s, s + R_n]$ and $[s + w_n, s + w_n + R_n]$. The interval between them is $(s + R_n, s + w_n)$. Its right end is X_F with $F = \{i < n : \varepsilon_i = 1\} \cup \{n\}$ and its length is g_n . Every finite nonempty F arises once. The total length is E minus the measure of $\mathcal{A}$, which is 1 [23, Section 7]. The series for g_n follows from $w_n = \sum_{j \geq 1} 2^{-jn}$ and $R_n = \sum_{j \geq 1} 2^{-jn} / (2^j - 1)$. $\square$

Lemma 2.2 (translation by a finite subsum). *Let F be finite with largest element n and let $0 \leq x \leq R_n$. The greedy rule applied to $X_F + x$ selects exactly F among the indices up to n and then agrees with the greedy rule applied to x . In particular $X_F + x \in \mathcal{A}$ if and only if $x \in \mathcal{A}$, and $X_F + x$ is rejected at a step $m > n$ exactly when x is.*

Proof. At an index $i \leq n$ the remainder is $X_{F \cap [i, n]} + x$. If $i \in F$ it is at least w_i and the index is taken. If $i \notin F$ it is at most $R_i - R_n + x \leq R_i < w_i$, so the index is skipped and nothing is rejected. After index n the remainder is x . $\square$

Two consequences. The map $x \mapsto x + 1$ preserves reduced denominators, so at every step $n \geq 2$ the number of fractions of height at most Q rejected at that step is even; every saved table has this property. The condition is $x \leq R_{\max F}$. It is not enough that $X_F + x \leq E$: $x = 1/2$ is not rejected through step 160, while $1/2 + 1/3 = 5/6$ lies in $(R_1, 1)$ and is rejected at step 1. Counts of rejections should therefore be taken modulo these translations. At height 200 the four fractions rejected at step 12 are $46/183$ and its translates by $1/3, 1$ and $4/3$, one event.

A finite subsum has odd denominator, since $\prod_{n \in F} (2^n - 1)$ is odd. So a rational with even denominator is never a finite subsum, and if it is a subsum at all its set S is infinite.

3 Negative results in the companion notes

Status is as stated in each note: L for checked in Lean there, O for an ordinary proof there, C for cited there.

Note	Location	Statement
#68 [19]	Section 6	Under the displayed cancellation hypotheses, O the integer-gap comparison fails at cutoffs $N = D + O(1)$ as the cancellation cutoff $D \rightarrow \infty$. Small tails alone do not give the strict comparison.

#243 [20]	Proposition 17	A counterexample has errors that are eventually nonzero, relatively small, with unbounded negative parts. This necessary profile is formalised; the comparison with scalar profiles and the need for denominator compatibility are ordinary discussion.	L, O
#249 [21]	Section 5	A rational series with the totient's values at odd indices, within 2 at even indices, and sum $5/4$. Positive tail differences need not be non-integral.	L
#249 [21]	Theorem 9	Every admissible rank-one quotient stays more than $21/320$ from its target.	L
#251 [22]	Proposition 1, Corollary 2	For every allowance $f(n) \rightarrow \infty$, sparse non-negative corrections reach every real in an interval while every fixed modulus eventually divides both the corrections and their cumulative sums. A bounded allowance is impossible under these congruences. Sources: [6], [7, Lemma 4], [8, Lemma 5.1], [9].	O
#257 [23]	Section 6	The small-displacement quantity stays above $1/2$ at full support, where the value is irrational [3]. No proof covering full support can rest on it.	L, O
#257 [23]	Section 3	Every positive divisor cover costs at least e times the mean of $\log^+$ of its multiplicity. The averaging method cannot reach the prime support, where irrationality is known at base 2 [11].	O
#269 [24]	Theorem 1	Nonsingular minors of every order: no finite sum of products separates one exponent from the other two. Fan posted the two-prime separation [12]; the three-prime statement is the note's.	L
#1049 [25]	Theorem 5	One family of nonzero integer-polynomial linear forms with common leading degree, coefficient-height and decay bounds at every fixed real $x > 1$ has $\sigma \leq \delta$; the sufficient cut-off $\sigma/(\sigma + \delta)$ supplied by those estimates is at most $1/2$. Ordinary proof, with the contradiction step and the comparison with $1/2$ checked in Lean.	O, L

#1049 [25]	Theorem 7	The stated clearing conditions cannot be met at base 3/2.	L
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Theorem 1.1(ii) explains interval filling in the #251 construction. The #249 counter-model is a separate explicit construction with its own preserved identities. The #1049 restriction does not exclude stronger base-specific estimates, different families at different bases, or forms involving several target values. The others bound a method. We tried to state one inequality that covers #1049 Theorem 5 and the cover cost of #257, a cost of clearing denominators against the decay gained, and did not find a formulation that survives both sets of hypotheses. We do not claim the rows share a cause.

4 The computation and its limits

The computation [2] runs the greedy rule in exact arithmetic on every reduced fraction in $(0, E]$ with $2 \leq q \leq Q$.

First reading. At $Q = 36$, 382 of 633 fractions are not rejected through step 160 and 14 are finite subsums, a share $(382 + 14)/633 = 0.6256$ close to $1/E = 0.6224$. This was read as evidence that about 62% of rationals are subsums, hence that #257 is false. The reading is wrong. By Theorem 1.4 of the note the share at any fixed depth tends to $2^N R_N/E$ whatever the truth of #257, because fractions equidistribute over the 2^N intervals that survive N steps.

An incorrect stopping rule. A subsequent interpretation went too far in the opposite direction, asserting that survival after about $2 \log_2 Q - 3.3$ steps was forced. The measure-based main term for the number rejected at step n is $2^{n-1} g_n \sum_{2 \leq q \leq Q} \varphi(q)$, asymptotically $(3Q^2/\pi^2)2^{n-1} g_n$, which is below 1 for $n > 2 \log_2 Q - 2 \log_2 \pi$, about step 12 at $Q = 200$. A main term below 1 does not make the actual count zero. Theorem 1.4 is a fixed-depth asymptotic, and its error $O(2^N \log Q/Q)$ does not justify an extrapolation to $N \sim 2 \log_2 Q$. Late rejections remain exact nonmembership certificates; the 12,218 fractions not rejected through step 60 have only that finite-depth status. At $N = 12$ the observed share is 0.62372 against $2^{12} R_{12}/E = 0.62245$.

An exact rejection at step 17. The witness 189/388, recorded in the earlier investigation's Desk B report, contradicts the proposed stopping rule: $2 \log_2 388 - 2 \log_2 \pi$ is about 13.9. The selected indices through step 16 are $F = \{2, 3, 7, 9, 10, 14, 15, 16\}$. At each skipped earlier index n , exact rational arithmetic gives a remainder at most $2^{-n} < R_n$, so no rejection has yet occurred. The remaining value is

$$r = \frac{189}{388} - X_F = \frac{9291822600689}{1217890317075045460}.$$

Since $1/(2^k - 1) = 2^{-k} + 4^{-k}/(1 - 2^{-k})$ and $1/(1 - 2^{-k}) \leq 2$ for $k \geq 1$, summing gives

$$R_n \leq 2^{-n} + \frac{2}{3 \cdot 4^n}.$$

At $n = 17$, the exact comparison is

$$R_{17} \leq \frac{196609}{25769803776} < r < \frac{1}{131071} = w_{17}.$$

The same rejection occurs for $577/388 = 1 + 189/388$ by Lemma 2.2. The independent reproduction, including every earlier skipped step, is in `research/experiments/sparse_interpolation/late_rejection.py`. This certificate shows that deeper computation can add exclusions. It does not convert survival to any finite depth into a membership certificate.

What survives is the agreement itself. At $Q = 200$ the counts at steps 1 to 9 are 4809, 1470, 600, 268, 132, 66, 32, 8, 6, against 4811, 1467, 604, 277, 133, 65, 32, 16, 8 from the measures of Lemma 2.1; steps 10 and 11 have none against 4 and 2, and step 12 has 4 against 1. The late counts fluctuate more than independent events would, because rejections arrive in the families of Lemma 2.2.

5 Eliminated ideas

1. *If the reachable values form a null set, no rational value is reachable.* False. Take the binary series with digit 1 everywhere except digit 0 at positions $n_1 < n_2 < \dots$, and allow each of those digits to be changed to 1. The reachable values form a null set when the positions are sparse, by Theorem 1.1(i), yet changing all of them gives $\sum 2^{-n} = 1$. So no count of choices against contraction, and no depth depending only on sparsity, excludes a particular rational.
2. *Every infinite subset of a host with null subsum set has an irrational sum.* A host is a set B of allowed indices, and its subsum set is $\{X_S(2) : S \subseteq B\}$. For hosts chosen without reference to the target this is open and is a form of #257 itself. As a universal statement it cannot be a route: the support of any rational subsum with infinite S would be such a host. The subsum set of a host B has positive measure exactly when the complement of B is finite, by Theorem 1.1(i) and the measure at full support.
3. *The share of surviving fractions as evidence.* Section 4.
4. *A wrong locator.* A draft of Theorem 1.3(b) cited Theorem 1.1 of [15] as Nishioka's theorem. That theorem says a value of a Mahler function at an algebraic point is rational or transcendental, which cannot prove irrationality. The proof in the note uses Nishioka's value theorem as quoted in [14, Theorem 1.1, p. 3], applied there to the two-dimensional system for $(g, 1)$ with regular points in $(0, 1)$.
5. *Algebraic independence for #1049.* With g as in the note, $\sum_{n \geq 1} (t^n - 1)^{-1} = \sum_{m \text{ odd}} g(t^{-m})$, and each $g(t^{-m})$ is transcendental for rational $t > 1$. This gives nothing for the infinite sum: limits of transcendental numbers take every value. No applicable value theorem for this decomposition is supplied here.

6 What was already known

- The covering argument of Theorem 1.1(ii) goes back to Kakeya; see [4, 5, 6]. Its use to build rational series inside a class defined by soft data is the method of Kovač and Tao [8], of Crmarić and Kovač [7] and of van Doorn and Kovač [9].
- That the subsums of $\sum (t^n - 1)^{-1}$ form a Cantor set at fixed $t \geq 2$ is [8, Remark 4.1].

- A closed set of positive measure can contain essentially no rationals [16], so the heuristic of the note's Section 4 cannot be a consequence of measure.
- The rational-point counting papers examined here concern null Cantor sets [17, 18]. We did not locate a theorem settling the present positive-measure subsum problem. The searches were made on 20 September 2026 and are listed in the repository record.
- One identity with a consequence. Since $\sum_{n \geq 1} \mu(n)/(b^n - 1) = 1/b$, the sums of $(b^n - 1)^{-1}$ over squarefree n with an even, respectively odd, number of prime factors are $\frac{1}{2}(X_{\text{sf}}(b) \pm 1/b)$. Duverney and Tachiya prove that $X_{\text{sf}}(2^j)$ is irrational [10, Corollary 1.2 and Example 1.1], as quoted in [23, Section 6], so both sums are irrational at every base 2^j . Both supports have divergent reciprocal sums. The identity at base 2 is derived in [21, Section 5]; the corollary may be known.

7 Questions

1. Is $1/2$ a subsum of $\sum (2^n - 1)^{-1}$? The exact obligation is in [23, Theorem 7]. By Lemma 2.1 the general question is one-sided approximation of a rational by finite subsums X_F to within $g_{\max F}$.
2. Does the count of fractions of height at most Q rejected at step n stay close to $(3Q^2/\pi^2)2^{n-1}g_n$ in the joint range $n \leq (2 - \varepsilon)\log_2 Q$, counted modulo the translations of Lemma 2.2? A persistent excess would be the first sign of an arithmetic mechanism for #257.
3. Is $a^2 > b^3$ necessary in Theorem 1.3(a)? First case: base $4/3$, a chain with ratios 2 and infinitely many ratios 3.
4. Is there one inequality behind #1049 Theorem 5 and the cover cost of #257?
5. For factorial denominators, what changes when every fixed modulus must eventually divide both the corrections and their cumulative sums? Without those congruences, the allowance $F(n) = n-1$ for $n \geq 2$ gives the full interval $[0, 1]$, whereas $F(n) = o(n)$ as $n \rightarrow \infty$ gives a null attainable set. The latter conclusion does not follow from small allowances merely along a subsequence, as the example below shows.

The final question, including arbitrary oscillating allowances, is answered in [An exact capacity criterion for series with eventual congruences](#). For positive integer denominators with $Q_n \mid Q_{n+1}$ and $Q_{n+1} \geq 2Q_n$, and integer allowances $F_n \geq 0$ with $\sum F_n/Q_n < \infty$, eventual allowance and eventual divisibility of every digit by each fixed integer represent every nonnegative real exactly when $Q_N \sum_{n > N} F_n/Q_n \rightarrow \infty$; otherwise the attainable set is null and meagre. In the positive case the cumulative congruences also hold, with cutoffs independent of the target. The factorial support-gap threshold is a corollary. This is an ordinary theorem of that note; the complete capacity criterion is not claimed as Lean-checked.

For the nullity assertion, eventually $F(n) + 1 \leq n/2$ and $F(n) \leq n$. Thus the number of prefixes through N is at most $C2^{-N}N!$ for a fixed C , while the capacity after N is at

most $\sum_{n>N} n/n! \leq 2/N!$. The covering bound of Theorem 1.1(i), allowing zero-capacity levels to be omitted, tends to zero.

For the subsequence counterexample put $F(2k) = 0$ and $F(2k+1) = (2k)(2k+1) - 1$ for $k \geq 1$, with $F(1) = 0$. Then

$$\frac{F(2k+1)}{(2k+1)!} = \frac{1}{(2k-1)!} - \frac{1}{(2k+1)!}.$$

The total capacity is 1, and the capacity after each permitted index $2k+1$ is exactly $1/(2k+1)!$, equal to the spacing between its choices. The interval criterion therefore gives every value in $[0, 1]$, even though $F(n)/n = 0$ at every even index. The exact telescoping identities and sample greedy expansions are reproduced by the script cited in Section 4; the interval conclusion follows from this argument, not from the samples.

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